

```
proc import datafile="/home/u64384583/CRC_RAW.xlsx"
  OUT=WORK.CRC_RAW
  DBMS=xlsx
  REPLACE;
  GETNAMES=YES;
RUN;

/* QC ANALYSIS */

proc contents data=work.crc_raw varnum;
  title "CRC_RAW DATA OVERVIEW";
RUN;

proc print data=work.crc_raw (obs=15);
  title "CRC_RAW DATA OBSERVATION";
RUN;

PROC MEANS DATA= Work.CRC_RAW N MEAN MAX MIN NMISS;
  title "CRC_RAW DATA NUMERIC SUMMARY";
RUN;

PROC FREQ data=WORK.crc_raw;
  TABLES Sex 'APC Groups'n 'Cancer Type Detailed'n Carcinomatosis;
run;

proc sql;
select count (*) as total ,
count(distinct "Patient ID"n) as UNIQUE_PATIENTS
from Work.CRC_RAW;
quit;

proc sort data=work.crc_raw out=work.crc_raw_nodup nodupkey;
by "Patient ID"n;
run;

data work.crc_clean;
set work.crc_raw_nodup;
run;

/* DEMOGRAPIC DOMAIN */

data WORK.dm;
  set work.crc_clean(rename=(Sex=Sex_raw));

  attrib
    STUDYID length=$20 label="Study Identifier"
    DOMAIN length=$2 label="Domain Abbreviation"
    USUBJID length=$30 label="Unique Subject Identifier"
    SUBJID length=$15 LABEL="Subject Identifier"
    AGE length=8 LABEL="Age"
```

```
AGEU    LENGTH=$5    LABEL="Age Units"
SEX      LENGTH=$1    LABEL="Sex"
RACE     LENGTH=$20   LABEL="Race"
ETHNIC   LENGTH=$30   LABEL="Ethnicity"
COUNTRY  length=$20   LABEL="Country"
ARM       length=$40  label="Description of planned arm"
ARMCD    LENGTH=$20   LABEL="Planned arm code";
```

```
STUDYID = "crc_apc_impact_2020";
DOMAIN = "DM";
COUNTRY="USA";
RACE = "NOT REPORTED";
AGEU = "YEARS";
ETHNIC="NOT REPORTED";
SUBJID =strip('Patient ID'n);
USUBJID=cats(STUDYID,"-",SUBJID);
AGE='Age at Metastases'n;
```

```
if upcase(strip(Sex_raw))="MALE"
  THEN SEX = "M";
ELSE IF UPCASE (strip(sex_raw))="FEMALE"
  THEN SEX = "F";
ELSE SEX = "U";
```

```
ARM    = "NOT ASSIGNED";
ARMCD  = "NOTASSGN";
```

```
KEEP
STUDYID DOMAIN USUBJID SUBJID
AGE AGEU SEX RACE ETHNIC
COUNTRY ARM ARMCD;
```

```
RUN;
```

```
PROC PRINT DATA=WORK.dm (obs=10);
RUN;
```

```
proc contents data=WORK.dm varnum;
  title "work.dm OVERVIEW";
RUN;
```

```
proc means data=WORK.dm n mean max min nmiss;
  title "work.dm summary";
run;
```

```
proc freq data=work.dm;
  tables  SEX RACE  ;
run;
```

```
/* MEDICAL HISTORY DOMAIN */
```

```
data work.mh;

    set work.crc_clean;

    attrib
        STUDYID length=$20 label="Study Identifier"
        DOMAIN length=$2 label="Domain Abbreviation"
        SUBJID length=$15 LABEL="Subject Identifier"
        USUBJID length=$30 label="Unique Subject Identifier"
        MHSEQ length=8 label="Sequence Number"
        MHTERM length=$100 label="Reported Medical History Term"
        MHCAT length=$40 label="Category for Medical History"
        MHSCAT length=$40 label="Subcategory for Medical History";

    STUDYID = "crc_apc_impact_2020";
    DOMAIN = "MH";
    SUBJID = strip('Patient ID'n);
    USUBJID=cats(STUDYID,"-",SUBJID);
    MHSEQ = _N_;
    MHTERM = "Cancer Type"n;
    MHCAT = "ONCOLOGY";
    MHSCAT = "Cancer Type Detailed"n;

    keep
        STUDYID DOMAIN USUBJID SUBJID MHSEQ MHTERM MHCAT MHSCAT;

run;
PROC PRINT DATA=work.mh;
RUN;

proc contents data=work.mh varnum;
    title "work.mh overview";
run;

proc means data=work.mh n mean max min nmiss;
    title "work.mh summary";
run;

proc freq data=work.mh;
    tables MHTERM MHCAT MHSCAT;
run;

/* TUMOR IDENTIFICATION DOMAIN */

data work.tu;

    set work.crc_clean;
```

```
attrib
```

```
STUDYID length=$20 label="Study Identifier"  
DOMAIN length=$2 label="Domain Abbreviation"  
USUBJID length=$30 label="Unique Subject Identifier"  
SUBJID length=$15 LABEL="Subject Identifier"  
TUSEQ length=8 label="Sequence Number"  
TULNKID length=$20 label="Tumor Link Identifier"  
TUTESTCD length=$8 label="Tumor Test Short Name"  
TUTEST length=$40 label="Tumor Identification Test"  
TUORRES length=$100 label="Result of Tumor Identification"  
TUCAT length=$40 label="Tumor Category";
```

```
STUDYID = "crc_apc_impact_2020";  
DOMAIN = "TU";  
SUBJID = strip(put('Patient ID'n,$15.));  
USUBJID = cats(STUDYID,"-",SUBJID);  
TUSEQ = _N_;  
TULNKID = cats("TUMOR-",TUSEQ);  
TUTESTCD = "TUMIDENT";  
TUTEST = "Tumor Identification";  
TUORRES = strip('Cancer Type Detailed'n);  
TUCAT = strip('Cancer Type'n);
```

```
keep
```

```
STUDYID  
DOMAIN  
USUBJID  
SUBJID  
TUSEQ  
TULNKID  
TUTESTCD  
TUTEST  
TUORRES  
TUCAT;
```

```
run;
```

```
proc print data=work.tu(obs=10);  
title "TU DOMAIN";  
run;
```

```
proc contents data=work.tu varnum;  
title "work.tu overview";  
run;
```

```
proc means data=work.tu n mean max min nmiss;  
title "work.tu summary";  
run;
```

```
proc freq data=work.tu;  
tables TUORRES TUCAT;  
run;
```

```
/* CLINICAL EVENTS DOMAIN */
```

```
data work.ce;
```

```
set work.crc_clean;
```

```
attrib
```

```
STUDYID length=$20 label="Study Identifier"  
DOMAIN length=$2 label="Domain Abbreviation"  
USUBJID length=$30 label="Unique Subject Identifier"  
SUBJID length=$15 label="Subject Identifier"  
CESEQ length=8 label="Sequence Number"  
CETERM length=$100 label="Reported Clinical Event Term"  
CECAT length=$40 label="Category for Clinical Event"  
CEOCCUR length=$1 label="Clinical Event Occurrence";
```

```
STUDYID = "crc_apc_impact_2020";  
DOMAIN = "CE";  
SUBJID = strip(put('Patient ID'n,$15.));  
USUBJID = cats(STUDYID,"-",SUBJID);  
CESEQ = _N_;  
CETERM = "CARCINOMATOSIS";  
CECAT = strip('Cancer Type'n);
```

```
if upcase(strip(Carcinomatosis)) = "YES" then  
    CEOCCUR = "Y";
```

```
else if upcase(strip(Carcinomatosis)) = "NO" then  
    CEOCCUR = "N";
```

```
else CEOCCUR = "";
```

```
keep
```

```
STUDYID  
DOMAIN  
USUBJID  
SUBJID  
CESEQ  
CETERM  
CECAT  
CEOCCUR;
```

```
run;
```

```
proc print data=work.ce (obs=10);  
title "CE DOMAIN";  
run;
```

```
proc contents data=work.ce varnum;
```

```
title "work.ce overview";
run;

proc means data=work.ce n mean max min nmiss;
title "work.ce summary";
run;

proc freq data=work.ce;
tables CECAT;
run;

/*=====
      ADaM MUTATION ANALYSIS DATASET
      USING MERGE
=====*/

/* Sort Datasets */

proc sort data=work.dm;
by USUBJID;
run;

proc sort data=work.tu;
by USUBJID;
run;

/*=====
      CREATE ADMUT
=====*/

data work.admut;

merge
    work.dm(in=a)
    work.tu(in=b);

by USUBJID;

if a and b;

attrib
    PARAMCD length=$20 label="Parameter Short Name"
    PARAM length=$100 label="Parameter Description"

    GENE length=$20 label="Gene Name"
    MUTSTAT length=$20 label="Mutation Status"

    AVAL length=8 label="Analysis Value"
    AVALC length=$10 label="Analysis Value Character";
```

```
/* Mutation Parameter */

PARAMCD = "APCMUT";

PARAM   = "APC Mutation Analysis";

/* Gene */

GENE = "APC";

/* Mutation Status Logic */

if not missing(TUCAT) then do;

    MUTSTAT = "MUTATED";

    AVAL = 1;

    AVALC = "YES";

end;

else do;

    MUTSTAT = "WILDTYPE";

    AVAL = 0;

    AVALC = "NO";

end;

keep
    STUDYID
    USUBJID
    SUBJID

    AGE
    SEX

    PARAMCD
    PARAM

    GENE
    MUTSTAT

    AVAL
    AVALC

    TUORRES
    TUCAT;
```

```
run;
```

```
/*=====
                        QC CHECKS
=====*/
```

```
proc print data=work.admut(obs=10);
  title "ADMUT DATASET";
run;
```

```
proc contents data=work.admut varnum;
  title "ADMUT OVERVIEW";
run;
```

```
proc freq data=work.admut;
  tables
    GENE
    MUTSTAT
    TUCAT;
run;
```

```
proc means data=work.admut n mean min max;
  var AVAL AGE;
run;
```

```
proc freq data=work.admut;
  tables MUTSTAT;
  title "Mutation Frequency";
run;
```

```
proc freq data=work.admut;
  tables GENE * MUTSTAT;
  title "Gene vs Mutation Status";
run;
```

```
proc freq data=work.admut;
  tables TUCAT * MUTSTAT / chisq;
  title "Cancer Type vs Mutation";
run;
```

```
proc sql;
  select
    MUTSTAT,
    count(*) as COUNT,
    calculated COUNT /
    (select count(*) from work.admut)*100
  as PERCENT format=8.2
  from work.admut
  group by MUTSTAT;
quit;
```

```
/*=====
```

TLF REPORTS
TABLES LISTINGS FIGURES

=====*/

/*=====

TABLE 1
DEMOGRAPHIC SUMMARY

=====*/

title "Demographic Summary Table";

```
proc freq data=work.dm;  
  tables SEX RACE ETHNIC COUNTRY / nocum;  
run;
```

```
proc means data=work.dm n mean std min max;  
  var AGE;  
run;
```

/*=====

TABLE 2
MUTATION SUMMARY

=====*/

title "Mutation Summary Table";

```
proc freq data=work.admut;  
  tables MUTSTAT GENE / nocum;  
run;
```

/*=====

TABLE 3
CANCER TYPE VS MUTATION STATUS

=====*/

title "Cancer Type vs Mutation Status";

```
proc freq data=work.admut;  
  tables TUCAT * MUTSTAT / chisq norow nocol;  
run;
```

/*=====

LISTING 1
SUBJECT LISTING

=====*/

title "Subject Listing";

```
proc print data=work.dm(obs=20);
```

```
  var
```

```
USUBJID  
AGE  
SEX  
RACE  
COUNTRY;
```

```
run;
```

```
/*=====
```

```
LISTING 2
```

```
MUTATION LISTING
```

```
=====*/
```

```
title "Mutation Analysis Listing";
```

```
proc print data=work.admut(obs=20);
```

```
var
```

```
USUBJID
```

```
GENE
```

```
MUTSTAT
```

```
AVAL
```

```
TUORRES
```

```
TUCAT;
```

```
run;
```

```
/*=====
```

```
FIGURE 1
```

```
MUTATION STATUS BAR CHART
```

```
=====*/
```

```
title "Mutation Status Distribution";
```

```
proc sgplot data=work.admut;
```

```
vbar MUTSTAT /
```

```
datalabel;
```

```
run;
```

```
/*=====
```

```
FIGURE 2
```

```
CANCER TYPE DISTRIBUTION
```

```
=====*/
```

```
title "Cancer Type Distribution";
```

```
proc sgplot data=work.admut;
```

```
vbar TUCAT /
```

```
datalabel;
```

```
run;
```

```
/*=====
      FIGURE 3
    SEX DISTRIBUTION
=====*/
```

```
title "Sex Distribution";
```

```
proc sgplot data=work.dm;
```

```
    vbar SEX /
        datalabel;
```

```
run;
```

```
/*=====
      EXPORT PDF REPORT
=====*/
```

```
ods pdf file="/home/u64384583/PROJECT.pdf";
```

```
title "Clinical Mutation Analysis Report";
```

```
proc freq data=work.admut;
```

```
    tables MUTSTAT;
```

```
run;
```

```
proc sgplot data=work.admut;
```

```
    vbar MUTSTAT;
```

```
run;
```

```
title "Sex Distribution";
```

```
proc sgplot data=work.dm;
```

```
    vbar SEX /
        datalabel;
```

```
run;
```

```
title "Cancer Type Distribution";
```

```
proc sgplot data=work.admut;
```

```
    vbar TUCAT /
        datalabel;
```

```
run;
```

```
ods pdf close;
```

